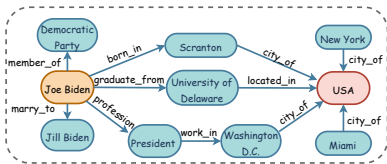


Knowledge Graphs (KGs)



Relation Paths $\mathcal{Z}$

z_1 : born_in $\rightarrow$ city_of
 z_2 : graduate_from $\rightarrow$ located_in
 z_3 : profession $\rightarrow$ work_in $\rightarrow$ city_of
...

Prompt: Please generate helpful relation paths for answering the question.

Question: What is the nationality of Joe Biden?

Inference

Instruction
Finetuning

$$Q(z|a, q, \mathcal{G})$$

Planning Optimization

$$P_{\theta}(z|q)$$

① Planning

② Retrieval

Joe Biden $\xrightarrow{\text{born_in}}$ Scranton $\xrightarrow{\text{city_of}}$ USA
Joe Biden $\xrightarrow{\text{graduate_from}}$ University of Delaware $\xrightarrow{\text{located_in}}$ USA
...

Prompt: Please answer the question based on the reasoning paths and explain why.

③ Reasoning

Answer: The answer is USA, because

$$P_{\theta}(a|q, \mathcal{Z}, \mathcal{G})$$

Reasoning Optimization